

Practical: 10

AIM: Capstone Project IOT Project.

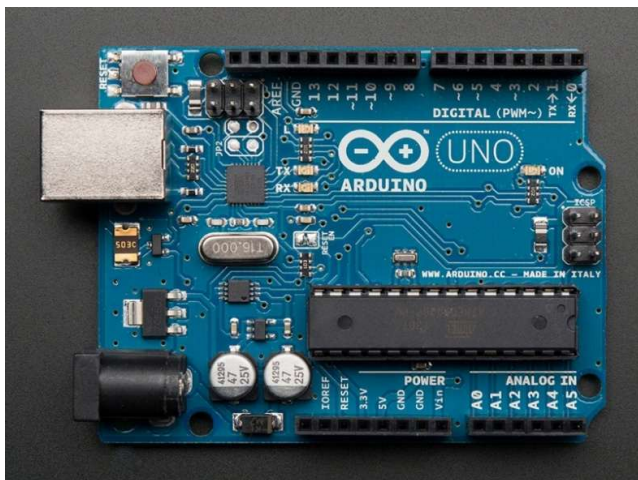
Project name: Solar sun tracking system:

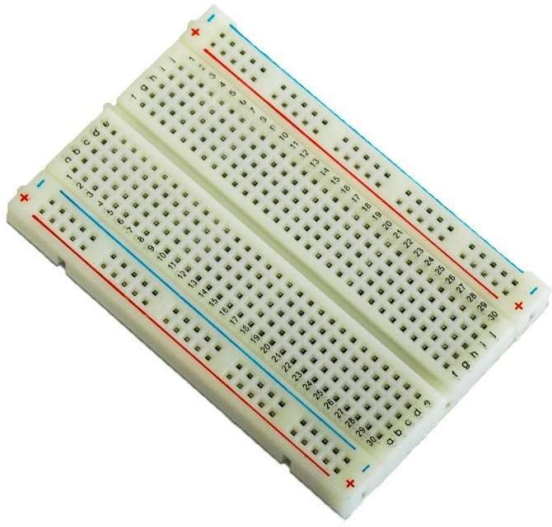
Aim: An automated system that aligns solar panels toward the sun using LDR sensors and a servo motor, maximizing energy efficiency and ensuring optimal sunlight capture throughout the day.

Component:

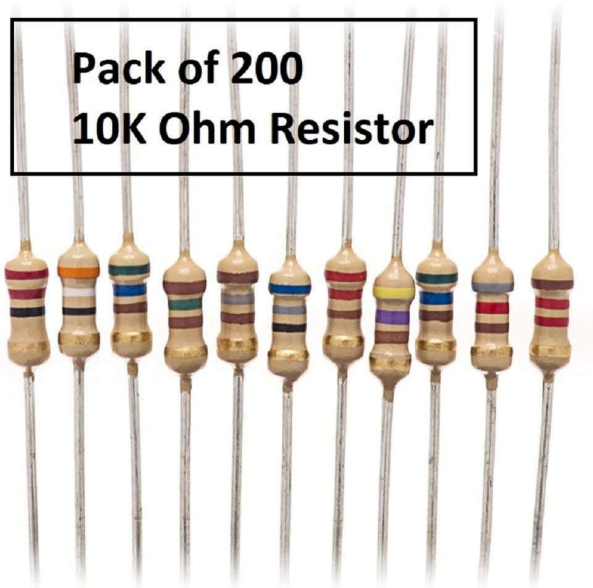
Name	Quantity	Component
S1	1	Solar
U1	1	Arduino uno
B1, B2	2	Breadboard
R1, R2, R3, R4, R5, R6	6	Register (220 and 10k)
L1, L2, L3, L4	4	LDR
M1	1	Servo motor
L1, L2, L3, L4	4	LED
J1.....Jn	-	Jumper wire

Component image:





**Pack of 200
10K Ohm Resistor**





Code:

```
#include <Servo.h>

Servo solarServo; // Servo object

// LDR pins
int LDR4 = A0; // Top Left
int LDR3 = A1; // Top Right
int LDR2 = A2; // Bottom Left
int LDR1 = A3; // Bottom Right

// Servo pin
int servoPin = 9;

// LED pins
int LED1 = 11; // Bottom Right LED
int LED2 = 10; // Bottom Left LED
int LED3 = 6; // Top Right LED
int LED4 = 3; // Top Left LED

// Variables
int servoPos = 90; // Initial servo position
int errorThreshold = 15; // Sensitivity threshold
unsigned long lastPrintTime = 0; // For 5-second Serial print delay

void setup() {
  Serial.begin(9600);
  solarServo.attach(servoPin);
  solarServo.write(servoPos);

  pinMode(LED1, OUTPUT);
  pinMode(LED2, OUTPUT);
  pinMode(LED3, OUTPUT);
  pinMode(LED4, OUTPUT);

  delay(1000);
}

void loop() {
  // --- Read LDR values ---
  int val4 = analogRead(LDR4); // Top Left
  int val3 = analogRead(LDR3); // Top Right
  int val2 = analogRead(LDR2); // Bottom Left
  int val1 = analogRead(LDR1); // Bottom Right

  // --- Calculate averages ---
  int topAvg = (val4 + val3) / 2;
  int bottomAvg = (val2 + val1) / 2;
  int leftAvg = (val4 + val2) / 2;
  int rightAvg = (val3 + val1) / 2;
```

```
// --- Differences for tracking ---
int verticalDiff = topAvg - bottomAvg;
int horizontalDiff = leftAvg - rightAvg;

// --- Smooth Servo Movement ---
if (abs(horizontalDiff) > errorThreshold) {
  if (horizontalDiff > 0 && servoPos > 0)
    servoPos--; // Move left
  else if (horizontalDiff < 0 && servoPos < 180)
    servoPos++; // Move right

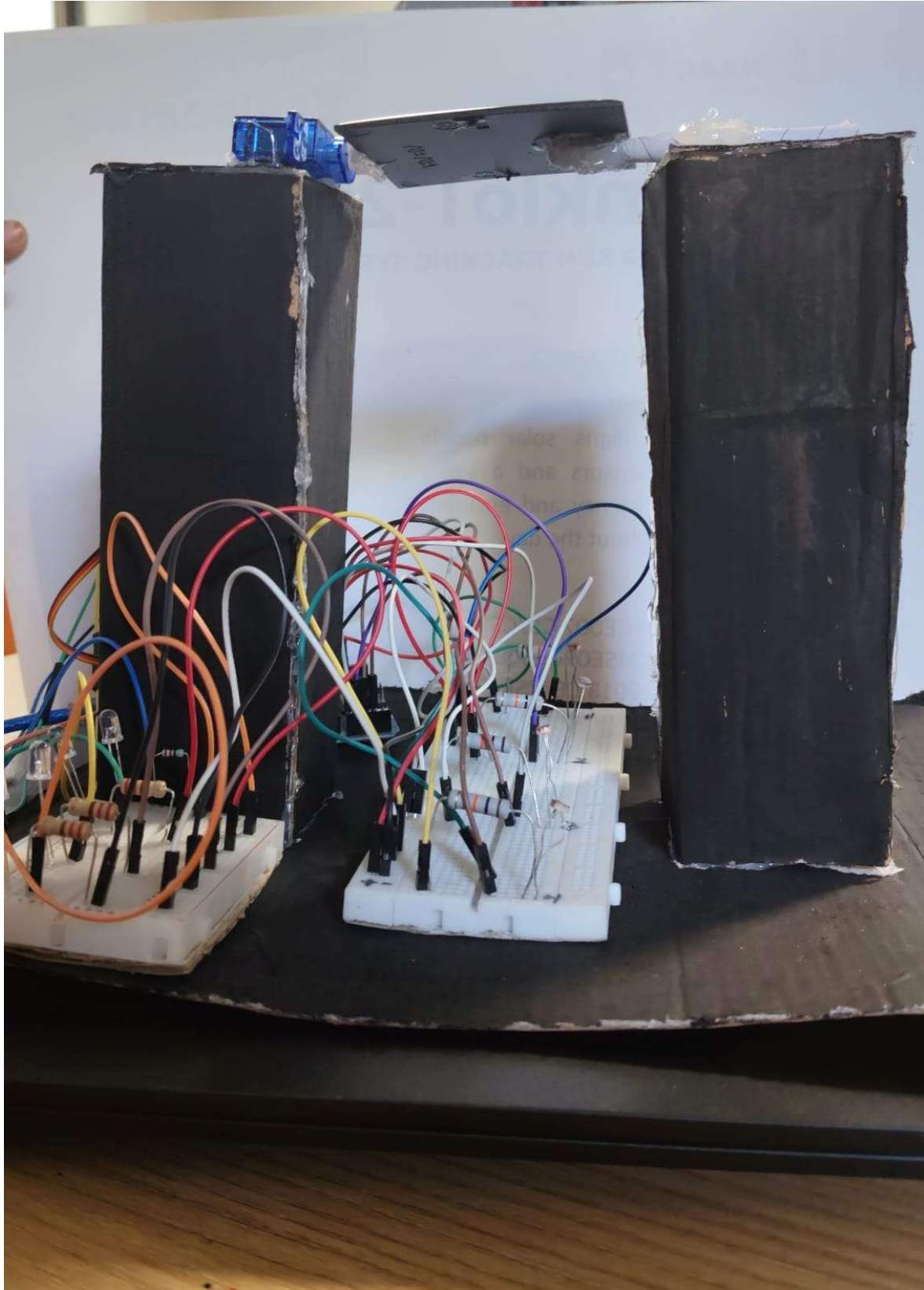
  solarServo.write(servoPos);
  delay(40); // smooth step delay
}

// --- LED Control: darkest LDR LED ON ---
int minLight = min(min(val1, val2), min(val3, val4));

// Turn OFF all LEDs first
digitalWrite(LED1, LOW);
digitalWrite(LED2, LOW);
digitalWrite(LED3, LOW);
digitalWrite(LED4, LOW);

// Light only the LED with the lowest (darkest) LDR value
if (minLight == val1) digitalWrite(LED1, HIGH); // Bottom Right
else if (minLight == val2) digitalWrite(LED2, HIGH); // Bottom Left
else if (minLight == val3) digitalWrite(LED3, HIGH); // Top Right
else if (minLight == val4) digitalWrite(LED4, HIGH); // Top Left

// --- Print every 5 seconds ---
if (millis() - lastPrintTime >= 5000) {
  lastPrintTime = millis();
  Serial.print("LDR1="); Serial.print(val4);
  Serial.print(" LDR2="); Serial.print(val3);
  Serial.print(" LDR3="); Serial.print(val2);
  Serial.print(" LDR4="); Serial.print(val1);
  Serial.print(" Servo="); Serial.println(servoPos);
}
}
```

Output:

Advantage:

1. Increases solar energy efficiency (25–40% more power).
2. Maximizes sunlight capture throughout the day.
3. Provides consistent and stable energy output.
4. Reduces required panel area for the same output.
5. Works effectively from sunrise to sunset.
6. Operates automatically with low maintenance.
7. Compatible with smart and IoT systems.
8. More cost-effective over long-term use.

Disadvantages:

1. Higher initial installation cost.
2. More complex design and setup.
3. Requires regular maintenance and calibration.
4. Mechanical parts may wear out over time.
5. Consumes a small amount of power for tracking.
6. Less efficient on cloudy or rainy days.
7. Not suitable for small-scale or low-budget systems.
8. Occupies more space due to movement range.

Conclusion:

A solar sun tracking system is an effective technology that significantly improves the efficiency of solar energy generation by continuously aligning solar panels with the sun's position. Although it involves higher initial cost and maintenance compared to fixed systems, its long-term benefits—such as increased power output, better energy utilization, and suitability for modern smart applications—make it a valuable solution for sustainable and efficient solar power generation.